

Federico P. Cortese, PhD

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Date of birth: January 1, 1994

[Google Scholar](#) - [GitHub](#) - [ORCID](#) - [ResearchGate](#)

RESEARCH INTERESTS

Latent variable models, multivariate time-series analysis, regime switching models, unsupervised learning

PROFESSIONAL POSITIONS

Research fellow (assegnista di ricerca)

Department of Economics, Management, and Quantitative Methods - University of Milan

Milan, Italy

July 2025 – Present

Researcher (ricercatore III livello)

National Research Council, Institute for Applied Mathematics and Information Technologies

Milan, Italy

October 2023 – June 2025

Visiting research fellow

Lund University

Lund, Sweden

May 2022 – October 2022

EDUCATION

PhD in Statistics

University of Milano-Bicocca

Milan, Italy

2019 - 2023

Thesis: “[Statistical Modeling and Temporal Clustering of Multivariate Time-Series with Applications to Financial Data](#)”

MSc in Finance and Insurance

University of Calabria

Rende, Italy

2017 - 2019

Thesis: “Tail Dependence in Financial Markets: A Dynamic Copula Approach”

Final grade: 110/110 cum laude

BSc in Mathematics

University of Calabria

Rende, Italy

2013 - 2017

Thesis: “Stochastic Differential Equations with Financial Applications” (in italian)

Final grade: 100/110

ARTICLES

Cortese, Federico P.; Pievatolo, Antonio

2026

[Spatio-Temporal Jump Model for Urban Thermal Comfort Monitoring](#)

Stochastic Environmental Research and Risk Assessment (SERRA)

Cortese, Federico P.; Kolm, Petter N.; Lindström, Erik

2026

[Generalized Information Criteria for High-Dimensional Sparse Statistical Jump Models](#)

AStA: Advances in Statistical Analysis

Cortese, Federico P.; Di Ruzza, Sara; Alessi, Elisa Maria

2025

[A statistical sparse jump model for automatic identification of dynamical transitions in the co-orbital regime](#)

Nonlinear Dynamics

Cortese, Federico P.; Pievatolo, Antonio

2025

[Statistical Jump Model for Mixed-Type Data with Missing Data Imputation](#)

Advances in Data Analysis and Classification

Cortese, Federico P.; Pennoni, Fulvia; Bartolucci, Francesco

2024

[Maximum likelihood estimation of regime switching Student-*t* copula model](#)

International Statistical Review

Cortese, Federico P.; Kolm, Petter N.; Lindström, Erik

2023

[What drives cryptocurrency returns? A sparse statistical jump model approach](#)

Digital Finance

Cortese, Federico P.

2019

[Tail dependence in financial markets: A dynamic copula approach](#)

Risks

PREPRINTS

- Cortese, Federico P.; Rossini, Luca 2025
[A comparison between initialization strategies for the infinite hidden Markov model](#)
- Cortese, Federico P.; Pievatolo, Antonio; Alessi, Elisa Maria 2025
[Fuzzy Jump Models for Soft and Hard Clustering of Multivariate Time Series Data](#)
- Lawuobahsumo Kokulo; Leccadito, Arturo; Cortese, Federico P.; Algieri, Bernardina 2024
[Investigating Time-Varying Correlations between Cryptocurrency and Financial Markets: A Gas-Based Approach](#)

SHORT PAPERS

- Cortese, Federico P.; Pievatolo, Antonio 2025
[Thermal Comfort Monitoring of Urban Areas: A Spatio-Temporal Jump Model Approach](#)
Innovation & Society - Statistics and Data Science for Evaluation and Quality
- Cortese, Federico P.; Pievatolo, Antonio; Alessi, Elisa Maria 2025
[Fuzzy Jump Model for Asteroids Co-orbital Regimes Identification](#)
Statistics for Innovation II - Springer Proceedings in Mathematics & Statistics
- Cortese, Federico P.; Pievatolo, Antonio 2024
[Modelling and forecasting correlated failure counts](#)
ICES 2024 - 2nd Italian Conference on Economic Statistics - Book of Short Papers
- Cortese, Federico P.; Kolm, Petter N.; Lindström, Erik 2023
[Generalized information criteria for sparse statistical jump models](#)
Symposium i anvendt statistik - Copenhagen Business School
- Cortese, Federico P.; Bartolucci, Francesco; Pennoni, Fulvia 2022
[A regime switching Student-t copula model for the analysis of cryptocurrencies data](#)
Book of Abstracts, Mathematical and Statistical Methods for Actuarial Sciences and Finance
- Bartolucci, Francesco; Pennoni, Fulvia; Cortese, Federico P. 2021
[Hidden Markov and regime switching copula models for state allocation in multiple time-series](#)
Book of abstracts and short papers, CLADAG, Firenze University Press

ONGOING PROJECTS

- Cortese, Federico P.; Farcomeni, Alessio
Robust Feature-Weighted Jump Models for Simultaneous Temporal Clustering and State-Conditional Variable Selection
- Leisen, Fabrizio; Cortese, Federico P.; Rossini, Luca
Copula-Based Compound Random Measures for the Modeling of Sparse Networks with Overlapping Community Structure
- Cortese, Federico P.; Rossini, Luca
Multivariate Quantile Regression through Infinite Hidden Markov Models
- Cortese, Federico P.; Alessi, Elisa Maria
rJM: An R Package for the Estimation of Statistical Jump Models of Various Types

TEACHING EXPERIENCE

- Academic year **2025-2026**
Lecturer – Statistics for Big Data for Economics and Business
BSc Economics and Management, University of Milan - 20 hours
Teaching assistant – Unsupervised Learning
MSc Data Science for Public Decision Making, University of Rome-Tor Vergata - 18 hours
- Academic year **2024-2025**
Teaching assistant – Unsupervised Learning
MSc Data Science for Public Decision Making, University of Rome-Tor Vergata - 12 hours
- Academic year **2022-2023**
Teaching assistant – Probability and Statistics
BSc Automation Engineering, Politecnico di Milano - 36 hours
Teaching assistant – Fundamentals of Statistics and Biomedical Signals
BSc Biomedical Engineering, Politecnico di Milano- 20 hours
Teaching assistant – Bayesian Inference
MSc Biostatistics, University of Milano-Bicocca- 12 hours
Tutor – Probability and Statistics for Computer Science
BSc Computer Science, University of Milano-Bicocca - 28 hours
- Academic year **2021-2022**
Teaching assistant – Fundamentals of Statistics and Biomedical Signals
BSc Biomedical Engineering, Politecnico di Milano- 20 hours
Teaching assistant – Bayesian Inference
MSc Biostatistics, University of Milano-Bicocca- 12 hours
- Academic year **2018-2019**
Tutor – Financial Econometrics
MSc Finance and Insurance, University of Calabria- 10 hours

THESIS SUPERVISION

- Simone Bosio - *University of Brescia - MSc in Economics* (co-supervised with prof. Enrico Ripamonti)

SOFTWARE

- [RSstcopula](#)
R/C++ implementation for estimation and simulation of regime-switching Student- t copula models.
- [GIC-for-SJM](#)
R/C++/Python implementation for estimation of Sparse Jump Models and model selection via Generalized Information Criteria.

RESEARCH PROJECTS

[LUMIO](#). Development of data analysis and machine-learning models to study asteroid dynamics and predict orbital evolution for the LUMIO mission.

[RAISE](#). Implementation of statistical models and data analysis strategies for urban thermal comfort within the Robotics and AI for Socio-economic Empowerment project, supported by the European Commission under NextGenerationEU.

URBANSENSE. Supervised and monitoring of the activities related the project – “Citizens as sensors: promoting inclusion through the development and application of a multifunctional wearable system for a citizen-oriented city”.

CONFERENCES

- ENBIS-26, 26th annual conference of the European Network for Business and Industrial Statistics - University of Florence - **Invited speaker**, September 7-9, 2026
- SIS-FENStatS 2026, 53rd SIS and 1st FENStatS Scientific Meetings, La Sapienza - University of Rome, - **Contributed Speaker**, June 22-25, 2026
- 5th ISI Regional Statistics Conference - Valletta, University of Malta - **Contributed Speaker**, June 3-5 2026
- CMStatistics2025, 19th International Conference of the ERCIM Working Group on Computational and Methodological Statistics and the network Computational and Financial Econometrics - Birkbeck, University of London - **Invited Speaker**, December 13-15 2025
- ISBIS 2025, International Society for Business and Industrial Statistics - University of Amsterdam - **Invited Speaker**, October 2-3 2025
- EPSC-DPS 2025, Europlanet Science Congress, Division for Planetary Sciences - Finlandia Hall, Helsinki - **Contributed Speaker**, September 7-12, 2025
- IES 2025, Innovation & Society: Statistics and Data Science for Evaluation and Quality - University of Padua, Brixen - **Invited Speaker** and **discussant**, June 25-27, 2025
- SIS 2025, Statistics for Innovation - University of Genoa - **Contributed Speaker**, June 16-18, 2025
- BISP14, Bayesian Inference in Stochastic Processes - National Research Council of Italy, Institute for Applied Mathematics and Information Technologies, Milan - **Chair**, May 26-28, 2025
- SDS, Statistics and Data Science - University of Milan and Politecnico di Milano, Italy - **Invited Speaker**, April 02-03, 2025
- Emerging Risks in Economics and Finance - University of Florence, Italy - **Invited Speaker**, September 12-13, 2024
- EcoSta 2023 - Waseda University, Tokyo, Japan - **Contributed Speaker**, August 1-3, 2023
- Statistics@Naples - University of Naples Federico II, Naples, Italy - **Contributed Speaker**, June 28-30, 2023
- CMStatistics2022, Computational and Methodological Statistics - King's College London, London, UK - **Invited Speaker**, December 17-19, 2022
- International FinTech Research Conference - Politecnico di Milano, Milan, Italy - **Contributed Speaker**, October 27-28, 2022
- MAF2022, Mathematical and Statistical Methods for Actuarial Sciences and Finance - University of Salerno, Fisciano (SA), Italy - **Contributed Speaker**, April 20-22, 2022
- CLADAG, 13th Scientific Meeting of the Classification and Data Analysis Group - University of Florence, Florence, Italy - **Participant**, September 9-11, 2021

WORKSHOPS

- “Advanced Time Series Methods” - University of Perugia, Perugia, Italy - Participant, June 19-20, 2023
- “Bayesian Nonparametrics for Complex Data” - University of Padua, Padua, Italy - Participant, January 24, 2020

OUTREACH ACTIVITY

Escape Moon. Educational activity developed within the [Co-Science](#) initiative to explain, through a game-based approach, the dynamics of celestial bodies and the Earth–Moon system to primary school children.

Space debris: what future for a sustainable use of near-Earth space? Public outreach event within the [Co-Science](#) initiative focused on the environmental challenges of space debris and the importance of sustainable practices in space exploration and satellite operations.

EDITORIAL ACTIVITY

Referee for *Statistics and Probability Letters* (Elsevier), *Nonlinear Dynamics* (Springer), *Statistical Modelling* (Sage Journals), *Metron* (Springer), *Discover Analytics* (Springer), *Advances in Computational Science and Engineering* (American Institute of Mathematical Sciences) and *Scientific Reports* (Nature).

TECHNICAL SKILLS

Proficient in R

Competent in Python

Familiar with MATLAB, C++, SAS

Softwares: L^AT_EX, R-Markdown, Git

Languages: Italian (native), English (fluent), French (basic)